Ques. no: 01

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Ans. to the ques. no: 02

There are different data attribute types. some of them are -

i) Ordinal attribute

ii) Nominal attribute

iii) Binary attribute

iv) Numeric attribute

Ordinal attribute : It represents rankings of data where the magnitude of successive values are not known.

Example: serial (1st, 2nd, 3rd. . . .), army rankings etc.

ii) Nominal attribute: This types of attribute represents categories, states on name of something.

Example: Hair-color = { red, brown, black, white}

iii) Binary attribute: It is a nominal attribute which has only 2 states. (0, 1).

Example: Gender (Male, Female).

iv) Numeric attribute: It is divided into two parts:

- Interval scaled: Values are scaled using equaly distribute unit.

  Example: ranges of age.

- Ratio scaled: It is a type of attribute that contain inherent zero point.

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Example : i) Ratio of water and oil.
ii) ~~Ratio of~~

Ans. to the ques. no: 03

Quartiles divides the sorted data into 4 equal parts. There are 3 quartiles:

i) Q_1 : 25% of data lies below it.

ii) Q_2 (median): 50% of data lies below it.

iii) Q_3 : 75% of data lies below it.

Why they are used:

Quartiles are used for data discovery. It is important because it measures the spread of data. As the data are sorted in quartiles, so understanding the data and finding hidden patterns and relationships between data

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becomes easy. For these reasons, quartiles is used in data discovery.

How they work:

Quartiles helps in data discovery by diving the data into 4 equal parts. so at a time we can work with one parts and then the others. Which makes work easy.

Ans. to the ques. no: 04

The qualities that define data quality is given below:

i) Data should be cleaned.

ii) Source of data should be trustworthy.

iii) Before using the data they should be preprocessed.

iv) Data should be collected and stored properly for using.

v) Any types of leakage and impure data should not be included or collected without checking.

These things matter because we use these data for different types of work. If the quality of these data are not proper then they

can lead various problems. such as: wrong prediction of any important task based on disqualified data.

Ans. to the ques. no: 1

Some sources of data impurities are mentioned below -

i) Dimensionality
ii) Noise
iii) Using wrong data format.
iv) unauthorized access
v) collecting data from unknown source.

Missing data means that in any dataset any attribute value is not included or collected. If the number of missing values is high then in prediction system the result can be biased. And if the ~~at~~ data of any import attribute is missing then it can lead to ~~unknown~~ problems because if any dataset is made to show something ~~important~~ and some datas are missing then the expected things can't be shown.

That's why missing data is not good. ~~Im~~ And necessary measures should be taken to solve this problem.

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